

Aim:- To design a database to create required tables for a college database using SQL.

Theory:- A database is an organized collection of unrelated data stored electronically in a computer system. A database Management System (DBMS) is software that allows users to create, manage & manipulate databases.

In Relational databases, data is stored in the form of tables (relation). Each table consists of:

Row (Record) - represent individual entries
Columns (Field / Attributes) - Represent properties of the data.

Key Concept:-

- Database schema: structure of database tables.
- Table: A collection of unrelated data organized into rows & columns.
- Data Types: Defines the types of data stored in columns such as INT, VARCHAR, DATE etc.

For example, in a college database, table may include

- Student
- Course
- Department

These table store information about students, course & departments.

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Database structure of College:-① Tbl-course

S.No.	Field Name	Datatype	length	Remark
1.	Course-id	Int	1	Primary Key, Auto Incr.
2.	Name	String	100	NOT NULL
3.	Reg-fee	Int	-	-
4.	Tution-fee	Int	-	-
5.	Exam-fee	Int	-	NOT NULL
6.	Active-status	Bool	-	Default 1

② Teacher - tbl

S.No.	Field Name	Datatype	length	Remark
1.	Teacher-id	Int	-	Primary Key, Auto Incr.
2.	Name	String	100	NOT NULL
3.	Gender	VARCHAR	10	male / female
4.	Qualification	VARCHAR	100	Degree
5.	Salary	INT	-	-
6.	Phone-No.	VARCHAR	15	-
7.	Gmail	VARCHAR	100	Unique
8.	Join Date	Date	-	Join Date
9.	Subject	VARCHAR	-	-
10.	Active-status	VARCHAR Bool	-	Default 1

(3) Subject-table

S.No.	Field Name	Data type	length	Remark
1.	Subject-id	INT	-	Primary key
2.	Name	VARCHAR	100	NOT null
3.	Subject-code	VARCHAR	100	Unique
4.	Course-id	INT	-	Foreign key
5.	Credits	INT	-	-
6.	Semester	INT	-	-
7.	Year	INT	-	-
8.	Active-status	Bool	1	Default = 1

(4) Subject table - teacher

S.No.	Field Name	Data type	length	Remark
1.	ST-id	INT	-	Primary key / Auto increment
2.	Subject id	INT	-	Foreign key
3.	Teacher id	INT	-	Foreign key
4.	Semester	INT	-	-
5.	Session	VARCHAR	20	-
6.	Class id	INT	-	Foreign key

(5) Tbl - exam held

S.No.	Field - Name	Data type	length	Remark
1	Exam held id	INT	-	Primary key / Auto incr.
2	Exam - id	INT	-	Foreign key
3	Exam date	Date	-	-
4	Sub id	INT	-	-
5.	Exame type	VARCHAR	50	Foreign key

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(6) Student-table

S.No.	Field Name	Datatype	Length	Remark
1.	Student-id	int		Primary Key / Auto Incr.
2.	Name	VARCHAR	100	Not Null
3.	Gender	VARCHAR	50	-
4.	DOB	Date	-	Not Null
5.	Courseid	int	-	Foreign Key
6.	Semester	int	-	-
7.	Address	VARCHAR	100	-
8.	Phone-no	VARCHAR	20	Not Null (Unique)
9.	Email	VARCHAR	100	Unique
10.	M-Name	VARCHAR	100	Not Null
11.	F-Name	VARCHAR	100	Not Null
12.	Add. Date	Date	-	Not Null
13.	Active-status	Bool	1	Default

(7) Exam-table

S.No.	Field-name	Datatype	Length	Remark
1	Exam-id	int		Primary Key / Auto incr.
2	Exam-Name	VARCHAR	100	Not Null
3	Subject-id	Int	-	Foreign Key
4	Course-id	Int	-	Foreign Key
5	Semester	int	-	-
6	Total marks	Int	-	Not Null
7	Passing marks	Int	-	-
8	Session	VARCHAR	20	-
9	Exam-mode	VARCHAR		

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8. Exam-Result table

S.No.	Field Name	Data type	length	Remark
1.	Result-id	int	-	Primary key / Auto incr
2.	Exam-id	int	-	Foreign key
3.	Student-id	int	-	Foreign key
4.	Total-marks	int	-	-
5.	Grade	VARCHAR	20	A/B/C/D/fail
6.	Remark	VARCHAR	100	optional
7.	Session	VARCHAR	20	-

9. Fee-table

S.No.	Field Name	Data type	length	Remark
1.	Fee-id	int	-	Primary key / Auto increment
2.	Student-id	int	-	Foreign key
3.	Deposite-date	Date	-	Not Null
4.	Amount	float	-	Not Null
5.	Mode	VARCHAR	50	Cash / UPI

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Procedure

1. Start . MySQL / SQL / Oracle
2. Create a new database
3. Use the database
4. Create tables using CREATE Table
5. Verify tables using DESCRIBE or SHOW Tables.

Program

Step 1 Create Database

```
CREATE DATABASE CollegeDB;
```

Step 2 Use database

```
USE CollegeDB;
```

Step 3 Create Department table

```
CREATE TABLE Department (  
    Dept_ID INT,  
    Dept_Name VARCHAR(50),  
    HOD VARCHAR(50));
```

Step 4 Create Student table

```
CREATE TABLE Student(  
    Student_id INT,  
    Student_name VARCHAR(50),  
    Age INT,  
    Dept_ID INT);
```

Step 5 Create Course Table

```
CREATE TABLE Course (  
    Course_id INT,  
    Course_Name VARCHAR(50),  
    Credits INT);
```

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OutputDepartment Table

Dept_id	Dept_Name	HOD
101	Computer Science	Dr. Sharma
102	Electrical	Dr. Verma
103	Mechanical	Dr. Singh

Student Table

Student_id	Student Name	Age	Dept_id
1	Rahul	20	101
2	Priya	19	102
3	Aman	21	103

Course Table

Course_id	Course Name	Credity
201	DBMS	4
202	Operating System	3
203	Computer Networks	3

Expt. No. _____

Page No. 7~~Step 6~~ : Display Table~~SHOW TABLES;~~~~Step 7~~ : View table structure;
~~DESCRIBE Student;~~Step 6 Insert data into Department table

INSERT INTO Department VALUES

(101, 'Computer Science', 'Dr. Sharma');

(102, 'Electrical', 'Dr. Verma');

(103, 'Mechanical', 'Dr. Singh');

Step 7 Insert data into student table

INSERT INTO Student VALUES

(1, 'Rahul', 20, 101),

(2, 'Priya', 19, 102),

(3, 'Aman', 21, 103);

Step 8 Insert data into Course Table

INSERT INTO Course VALUES

(201, 'DBMS', 4);

(202, 'Operating system', 3),

(203, 'Computer Science', 3);

Step 9

SELECT * FROM Department;

SELECT * FROM Student;

SELECT * FROM Course;

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Aim:- To create tables & apply constraints such as Primary key, Foreign key & NOT NULL in a database.

Theory:- In DBMS, constraints are rules applied to table columns to ensure data accuracy, validity & integrity. They restrict the types of data that can be inserted into a table.

Types of Constraints:-

1) Primary Key

- A primary key uniquely identifies each record in a table
- It cannot contain duplicate values
- It cannot contain NULL values

Ex Student_id IN T PRIMARY KEY

2) Foreign Key

- A Foreign key creates a relationship between two tables.
- It refers to the primary key of another table.
- It maintains referential integrity.

Ex FOREIGN KEY (Dept_id) REFERENCES Department (Dept_id)

3. NOT NULL :-

- The NOT NULL constraints ensures that a column cannot have NULL values.
- It Forces the user to enter a values

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Student_name VARCHAR(50) NOT NULL

These constraints helps maintain data consistency & relational integrity in a database.

Algorithm

1. Start the SQL environment (MySQL / Oracle / SQL)
2. Create a new database
3. Select the database using USE Command
4. Create a Department table with Primary key.
5. Create a Student table with PK, FK & Not Null constraints
6. Insert values into the tables
7. Display the records using SELECT Command.

Program

Step 1: Create Database

```
CREATE DATABASE CollegeDB;
```

Step 2: Select Database

```
USE CollegeDB;
```


output tableDepartment Table

Dept_ID	Dept_Name	HOD
101	Computer Science	Dr. Sharma
102	Electrical	Dr. Verma
103	Mechanical	Dr. Singh

Student Table

Student_ID	Student_Name	Age	Dept_ID
1	Rahul	20	101
2	Priya	19	102
3	Aman	21	103

Step 3: Create Department Table with constraints

```
CREATE TABLE Department(
    Dept_id INT PRIMARY KEY,
    Dept_Name VARCHAR(50) NOT NULL,
    HOD VARCHAR(50) NOT NULL);
```

Step 4: Create student Table with constraints

```
CREATE TABLE student(
    Student_id INT PRIMARY KEY,
    Student_Name VARCHAR(50) NOT NULL,
    Age INT,
    Dept_id INT,
    FOREIGN KEY (Dept_id) REFERENCES
        Department (Dept_id));
```

Step 5: Data insertion

Insert Data into Department table

```
INSERT INTO Department VALUES
(101, 'Computer Science', 'Dr. Sharma'),
(102, 'Electrical', 'Dr. Verma'),
(103, 'Mechanical', 'Dr. Singh');
```

INSERT INTO Students VALUES

```
(1, 'Rahul', 20, 101),
(2, 'Priya', 19, 102),
(3, 'Aman', 21, 103);
```


OutputDepartment

Dept. id	Dept. Name	HOD
101	Computer Science	Dr. Sharma
102	Electrical	Dr. Verma
103	Mechanical	Dr. Singh

Student

Student id	Student Name	Age	Dept. id
1	Rahul	20	101
2	Priya	19	102
3	Aman	21	103

Step 6:Display Record

```
SELECT * FROM Department;
SELECT * FROM Student;
```

Result - Thus, the Primary Key, Foreign Key & NOT NULL constraints were successfully applied to the tables in the data base.

Viva Questions:-

Q1 What is a primary key?

Ans A primary key uniquely identifies each record in a table & cannot contain NULL values.

Q2 What is Foreign key?

Ans A foreign key is used to create a relationship between two tables.

Q3 What is the purpose of NOT NULL constraint?

Ans It ensures that a column cannot store NULL values.

Q4 Can a table have more than one primary key?

Ans No, a table can have only one primary key.

Q5 What is Referential Integrity?

Ans It ensures that values in foreign key column match values in the referenced PK column. Teacher's Signature: _____